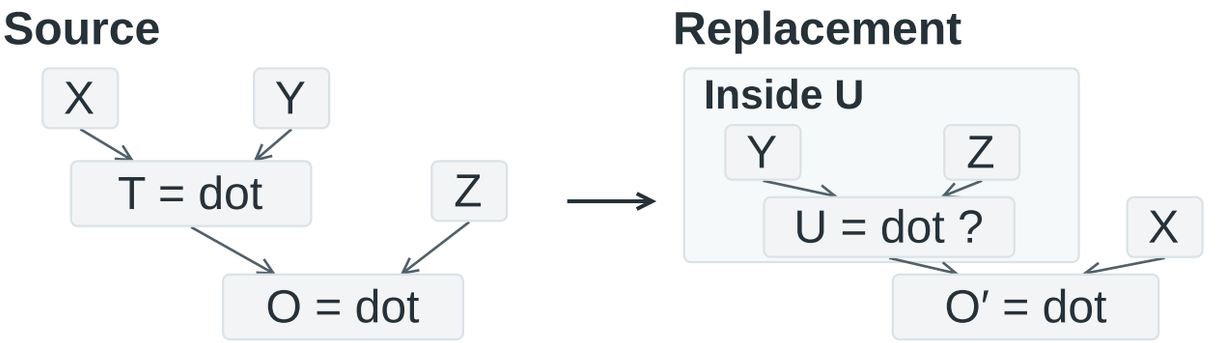



```
KeepAxes(U) = Axes_in(U) ∩
(Axes_out(U) ∪ Axes(O))
```

Input Y
{ K J_sel J_rem }

Input Z
{ I_sel J_sel F }

Axes inside U: Y ∪ Z
{ K I_sel J_sel J_rem F }

Needed outside U?
X: { K I_sel I_rem }

Yes → **KEEP**
Used by X
{ K I_sel }

Not needed here

Remaining candidates
{ J_sel J_rem F }

Required for output?
O: { I_rem J_rem F }

Yes → Required by O
{ J_rem F }

Not needed here

CONTRACT { J_sel }
Used only inside U

∪

Inferred roles in U = dot(Y, Z)

Left free	Contract	Right free
from Y	shared by Y, Z	from Z
{ K J_rem }	{ J_sel }	{ I_sel F }

I_rem arrives via X. Axis order is not shown.
No batch axes; position lists are constructed afterward.